

Jorge Arenas Amorós

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RESEARCH

Revisiting Fundamentals of the European Gas Market: The Role of Supply Substitution

- Develops a Bayesian SVAR with sign restriction over elasticities to identify European natural gas market drivers.
- Not all the supply shocks are alike: LNG supply disruptions have more permanent effects on prices than pipeline supply.
- Policy-relevant insights on energy security and market integration in the euro area.

New measures of market-based monetary policy surprises in macroeconomic outcomes

- Constructs a high-frequency monetary-policy-surprise measure repricing in contracts of Polymarket.
- Changes around discrete events (e.g., FOMC decisions, speeches, minutes) across different horizons.
- Monetary policy surprises are projected into different macro-financial time series using a monthly VAR.

Nonlinear Price Dynamics in Gas and Electricity Spot Markets

- Develops a nonlinear error-correction framework to jointly model electricity and natural gas spot prices.
- Smooth-transition dynamics to allow adjustment speeds and price pass-through to vary across regimes.
- Time-varying volatility and cross-market spillovers to address volatility clustering.

CONFERENCES

ICMAIF

International Conference on Macroeconomic Analysis and International Finance

May 2026

Rethymno, Greece

IAAE

International Association for Applied Econometrics Annual Conference

Jun. 2026

Lisbon, Portugal

EXPERIENCE

Teaching Assistant

University of Alicante — Fundamentos del Análisis Económico

2022 – present

Alicante, Spain

- *Introduction to Statistics*: probability theory and statistics (undergraduate)
- *Introduction to Macroeconomics*: introductory macroeconomics (undergraduate)
- *Mathematics I*: introductory algebra and calculus (undergraduate)
- *Statistics*: graduate course in statistics. Instructor: M. Dolores Collado
- *Econometrics*: graduate course in econometrics. Instructor: M. Dolores Collado

Data Replicator

Journal of the European Economic Association

2025 – present

Remote

- Check that data used during the replication package is cited appropriately.
- Ensured that code in replication packages for accepted papers ran correctly.
- Reproduce all the results from the manuscript.

EDUCATION

PhD in Economics

University of Alicante

Sep. 2023 – present

Alicante, Spain

- **Supervisor**: Iván Payá
- **Research**: Applied Macroeconomics, Time Series, Monetary Economics, Energy Economics.

Master in Quantitative Economics

University of Alicante

Sep. 2021 – June 2023

Alicante, Spain

- **Coursework**: Research-oriented master program, with a first year focused on core graduate courses in economics and a second year devoted to specialized elective coursework.
- **Master Thesis**: *Impact of Natural Gas and Crude Oil shocks on European stock returns*; analyzed the heterogeneous transmission of supply and demand shocks in natural gas and crude oil markets to European equity returns

Bachelor in Economics

Complutense University of Madrid

Sep. 2017 – June 2021

Madrid, Spain

- **Selected courses with honours**: Advanced Macroeconomics, International Finance, Regulation Economics, Macroeconomics III, Microeconomics II, Bachelor Thesis.
- **Last year specialization**: Economic Analysis mention.

ADDITIONAL TRAINING

Bayesian Models and Forecasting

Aug. 2024

Italian Econometrics Association

Venice, Italy

- Postgraduate course focused on Bayesian multivariate models and forecasting in economics and finance, including structural analysis, nonparametric methods, and forecast combinations, with hands-on MATLAB applications.
- Instructors: Marta Banbura, Roberto Casarin, Matteo Ciccarelli, and Francesco Ravazzolo

Financial Econometrics

Feb. 2025

University of Alicante

Alicante, Spain

- Graduate course on econometric methods for financial and macroeconomic time series, covering frequentist and Bayesian inference, estimation and testing with dependent observations, and applications including VaR/CoVaR, dependence modeling, and Bayesian SVARs.
- Instructor: Dante Amengual

Causal Inference with VARs

Nov. 2025

EABCN Training School

Online

- Covered VAR/SVAR foundations, identification challenges, and methods including short- and long-run restrictions, Cholesky decomposition, sign restrictions, heteroskedasticity-based identification, and proxy-SVAR approaches, with practical empirical sessions.
- Instructor: Giovanni Ricco

Energy and Commodity Prices: Macroeconomic Effects and Policy Implications

Apr. 2026

EABCN Training School

Vietri sul Mare, Italy

- Three-day course on the macroeconomics of energy and commodity prices, covering structural identification of energy price shocks in SVARs, energy price pass-through to inflation and monetary policy (local projections and proxy SVARs), and distributional dynamics (TVP-VAR, tail risks, and quantile/functional VAR approaches), with hands-on MATLAB sessions.
- Instructor: Hilde C. Bjørnland

Bayesian Analysis of Structural VAR Models

Jun. 2026

Lancaster University Management School

Lancaster, United Kingdom

- Three-day summer school on Bayesian analysis of structural VAR models, covering the identification problem, identification using inequality constraints, prior information about structural elasticities, and inference in set-identified models (credibility sets for IRFs, variance and historical decompositions), with hands-on MATLAB applications to oil supply and demand, monetary policy and labor market dynamics.
- Instructor: Christiane Baumeister

REFERENCES

- Iván Payá (University of Alicante)
- Ángel León (University of Alicante)

SKILLS

Programming : MATLAB, Python, R, Stata and SQL

Tools : GitHub, $\text{\LaTeX}$, Notion, Office

Languages: Spanish (native), English (fluent), Polish (beginner), Italian (beginner)